Week 10
Tuesday

07-04-2026

CVFA

★ **SEQUENCE:-** (CHAPTER 9 OF SECOND BOOK)

$2, 4, 6, \ldots$

$1, \frac{1}{2}, \frac{1}{4}, \ldots$

$2, P_1, P_2, 100, \ldots$

★ Correspondence

★ Def of Sequence = one-one (Injective) correspondence to a natural number is called sequence.

★ **SERIES:-**

Sum of the elements of sequence is called series.

★ **LIMIT POINT:-** (Kis point pe converge horaha hai).

$1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$

Here 0 is the limit point of this sequence

...neral term $= \{x_n\} = \dfrac{1}{2^{n-1}}$

This sequence is converging to '0'.

or

'0' is the limit point of the sequence.

$$\boxed{\lim_{n \to \infty} \frac{1}{2^{n-1}} = \frac{1}{2^{\infty - 1}} = \frac{1}{2^{\infty}} = \frac{1}{\infty} = 0}$$

when number comes = convergent sequence

x — x

1, 2, 3, 4, ...

$\{x_n\} = n$

$$\boxed{\lim_{n \to \infty} n = \infty}$$

divergent ~~series~~ sequence when '∞'

Harmonic Sequence x — x

$\rightarrow 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\ldots$

$\{x_n\} = \dfrac{1}{n}$

$$\boxed{\lim_{n \to \infty} \frac{1}{n} = \frac{1}{\infty} = 0}$$

CamScanner

* DEF :- "Let $\{x_n\}_{n=1}^{\infty}$ is a sequence.

'L' is called a limit point of the sequence if there exist $\epsilon > 0$ such that $|x_n - L| < \epsilon$; $\forall n \geq N$.

FOR EXAMPLE :

$$\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$$

$$\left|\frac{1}{n} - 0\right| < \epsilon$$

$$\left|\frac{1}{n}\right| < \epsilon$$

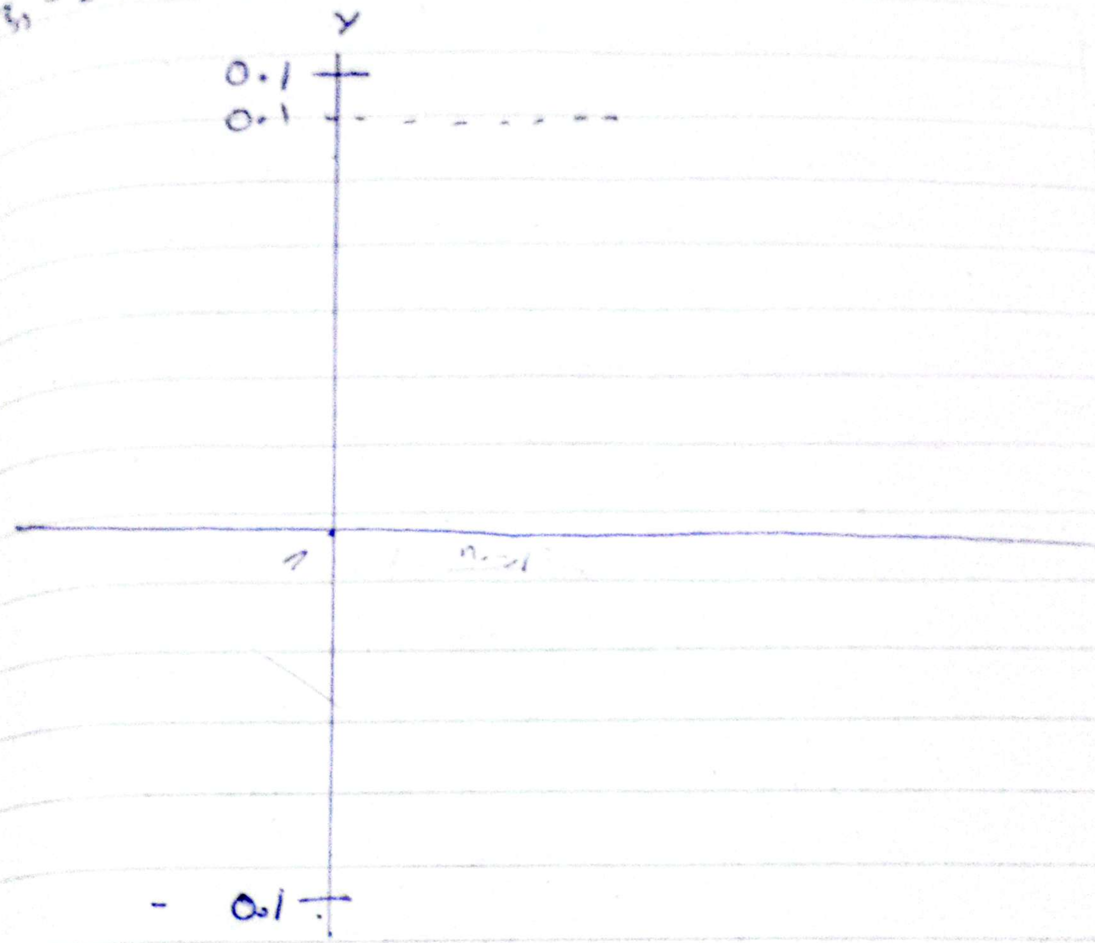
$$-\epsilon < \frac{1}{n} < \epsilon$$

Let

$\epsilon = 0.1$ open interval

$$-0.1 < \frac{1}{n} < 0.1$$

$|x_n - L| < \epsilon \; ; \; n \geq N$



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SECTION 9.3

* INFINITE GEOMETRIC SERIES:-

$$a + ar + ar^2 + \dots + ar^{n-1} + \dots \quad \text{—(1)}$$

Sum of n terms of Geometric Series:-

$$S_n = \frac{a(1-r^n)}{1-r}$$

CASE 1:-

* if $|r| < 1$:-

$$\lim_{n \to \infty} r^n = 0$$

R.w.e
$(0.1)^2 = 0.01$
$(0.1)^3 = 0.001$
r is decreasing

* $$S_\infty = \lim_{n \to \infty} S_n = \lim_{n \to \infty} \frac{a(1-r^n)}{1-r}$$

$$= \frac{a(1-0)}{1-r}$$

$$= \frac{a}{1-r}$$

Series 1 would be convergent as we will get a number (specific, defined.

* CASE #2 : IF $|r| > 1$

$$S_\infty = \frac{a(1-\infty)}{1-r} = \infty$$

This is divergent as ans is infinity.

* Ratio and Root test are not applied on negative term and they only tell nature not the sum.

Ex 9.3 Qs 3-14

Qs 4 $\sum_{k=1}^{\infty} \left(\frac{2}{3}\right)^{k+2} = \left(\frac{2}{3}\right)^3 + \left(\frac{2}{3}\right)^4 + \left(\frac{2}{3}\right)^5 \ldots$

$k=1$

Common ratio $= r = 2/3$

$|r| = 2/3 < 1$

$\therefore$ Series 1 is convergent series

$$S_\infty = \frac{a}{1-r} = \frac{(2/3)^3}{1-2/3} = \frac{(2/3)^3}{1/3} = \frac{8}{9}$$

Q.8 $\sum_{k=1}^{\infty}\left(\frac{1}{2^k}-\frac{1}{2^{k+1}}\right)$

$= \sum_{k=1}^{\infty}\frac{1}{2^k} - \sum_{k=1}^{\infty}\frac{1}{2^{k+1}}$

$= \left\{\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\dots\right\} - \left\{\frac{1}{2^2}+\frac{1}{2^3}+\frac{1}{2^4}+\dots\right.$

$r_1 = 1/2 \Rightarrow |r|<1$ $\qquad r_2 = \frac{1}{2}, |r|<1$

$= \left(\frac{a_1}{1-r_1}\right) - \left(\frac{a_2}{1-r_2}\right)$

$= \frac{1/2}{1-1/2} - \frac{(1/2)^2}{1-1/2} = 1-\frac{1}{2} = 1/2$

Application-

$$0.\overline{3} = 0.33333\ldots$$

$$= 0.03 + 0.03 + 0.003 + \ldots\ldots$$

$$\frac{0.03}{0.3} = \frac{0.003}{0.03} = 0.1$$

$$= \frac{a}{1-r}$$

$$= \frac{0.3}{1-0.1} = \frac{0.3}{0.9} = 1/3$$

10.131313

$= 10 + 0.\overline{13}$

$= 10 + (0.131313....)$

$= \cancel{10 + (0.13131313....})$

$= 10 + (0.13 + 0.0013 + 0.000013 + ...)$

Qs 21-24

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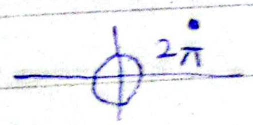
Chapter 11 :-

FOURIER SERIES :-

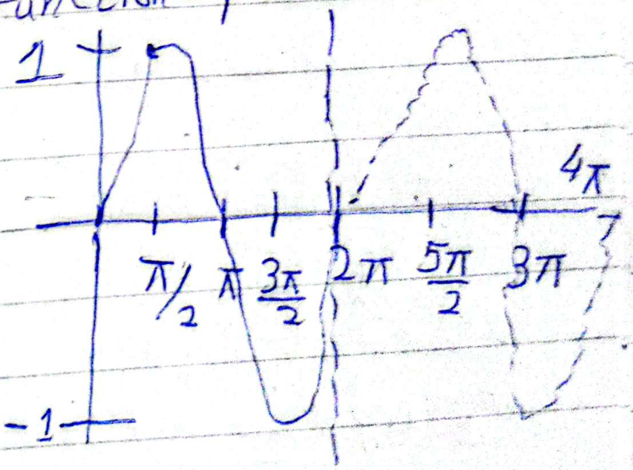
(i) Function must be periodic of period 2π

$$F(x+2\pi) = F(x)$$

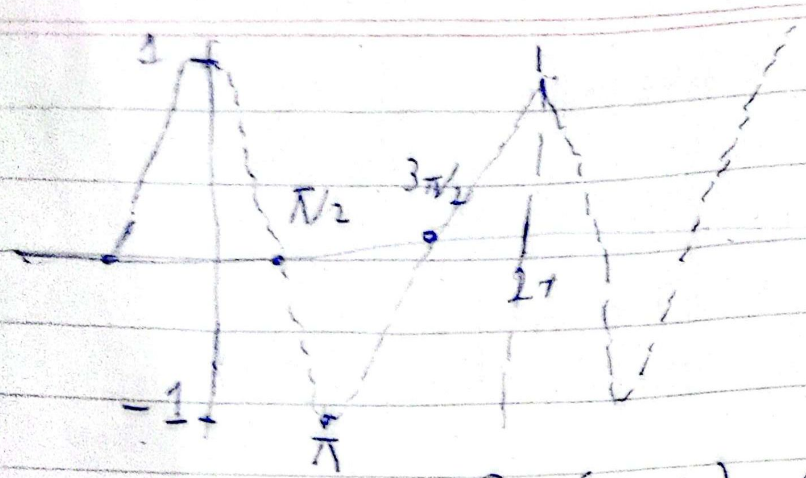


Eg. $\sin(x+2\pi) = \sin x$

* All trigonometric Function = periodic



Cos



$$\cos(\theta + 2\pi) = \cos\theta$$

* If $F(x)$ is a periodic function of period 2π then its Fourier series is defined as:

$$F(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos nx + b_n \sin x \right)$$

They are called Eulars Coefficient).

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(x)\,dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} F(x) \cos nx \, dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} F(x) \sin nx \, dx$$

- Fourier series is a type of convergent series.

$X - x$

Find the Fourier's Series For the Following function

(i) $F(x) = \begin{cases} -1 & -\pi < x < 0 \\ 1 & 0 < x < \pi \end{cases}$; $F(x+2\pi) = F(x)$

It indicates periodicity of period 2π. (Also called cyclic Function).

Required Fourier Series is:

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos nx + b_n \sin nx\right) \rightarrow (1)$$

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} F(x)\,dx = \frac{1}{2\pi}\left[\int_{-\pi}^{0}(-1)\,dx + \int_{0}^{\pi}(1)\,dx\right]$$

$$= \frac{1}{2\pi}\left[-x\Big|_{-\pi}^{0} + x\Big|_{0}^{\pi}\right]$$

$$-(0-(-\pi))$$

$$= \frac{1}{2\pi}\left[-(\cancel{-\pi-0}) + (\pi - 0)\right]$$

$$= \frac{1}{2\pi}\left[-\pi + \pi\right] = \boxed{0}$$

Note: Two coefficients can be zero but if you are getting all three zero you have made a mistake.

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} F(x)\cos nx\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-1)\cos nx\,dx + \int_{0}^{\pi}(1)\cos nx\,\cancel{dx}\right]$$

$$= \frac{1}{\pi}\left[\left\{-\frac{\sin nx}{n}\Big|_{-\pi}^{0}\right\} + \left\{\frac{\sin nx}{n}\Big|_{0}^{\pi}\right\}\right]$$

$$= \frac{1}{\pi}\left[-\frac{1}{n}\left\{\sin 0 - \sin n(-\pi)\right\} + \frac{1}{n}\left\{\sin n\pi - \sin 0\right\}\right]$$

$$a_n = \boxed{0}$$

$x \to \sin \to 0$

$y \to \cos \to 0$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} F(x)\sin nx\,dx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0}(-1)\sin nx\,dx + \int_{0}^{\pi}(1)\sin nx\,dx\right]$$

$$= \frac{1}{\pi}\left[\frac{\cos nx}{n}\Big|_{-\pi}^{0} + \left\{-\frac{\cos nx}{n}\Big|_{0}^{\pi}\right\}\right]$$

$$= \frac{1}{n\pi}\left[\{\cos 0 - \cos n(-\pi)\} - \{\cos n\pi - \cos 0\}\right]$$

$$= \frac{1}{n\pi}\left[(1-\cos n\pi) - (\cos n\pi - 1)\right]$$

$$\boxed{b_n = \frac{1}{n\pi}(2 - 2\cos n\pi)}$$

R.W

$\cos 1\pi = -1$
$\cos 2\pi = 1$
$\cos 3\pi = -1$
$\cos 4\pi = 1$
$\cos 5\pi = -1$
$\cos 6\pi = 1$
$\vdots$

$$= \frac{2}{n\pi}(1 - \cos n\pi)$$

$$b_n = \frac{2}{n\pi}\{1 - (-1)^n\}$$

*Putting all these coefficients in eqn ①

$$eq\ ① \Rightarrow F(x) = 0 + \sum_{n=1}^{\infty} (0)\cos nx + \frac{2}{n\pi}\{1-(-1)^n\}\sin nx$$

$$\begin{cases} -1 & -\pi < x < 0 \\ 1 & 0 < x < \pi \end{cases} = \frac{2}{\pi}\sum_{n=1}^{\infty}\left\{\frac{1-(-1)^n}{n}\right\}\sin nx$$

→ Answer

*Application of Fourier Series is converting piecewise function to continuous function

$$= \frac{2}{\pi}\left[\frac{1-(-1)^1}{1}\sin x + \frac{1-(-1)^2}{2}\sin 2x + \frac{1-(-1)^3}{3}\sin 3x + \frac{1-(-1)^4}{4}\sin 4x + \frac{1-(-1)^5}{5}\sin 5x + \dots\right]$$

$$F(x) = \frac{2}{\pi}\left[2\sin x + \frac{2}{3}\sin 3x + \frac{2}{5}\sin 5x + \frac{2}{7}\sin 7x\right]$$

$$\begin{cases} -1 & -\pi < x < 0 \\ 1 & 0 < x < \pi \end{cases} = \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\sin(2n-1)x}{(2n-1)}$$

1, 3, 5, 7, 9 ...

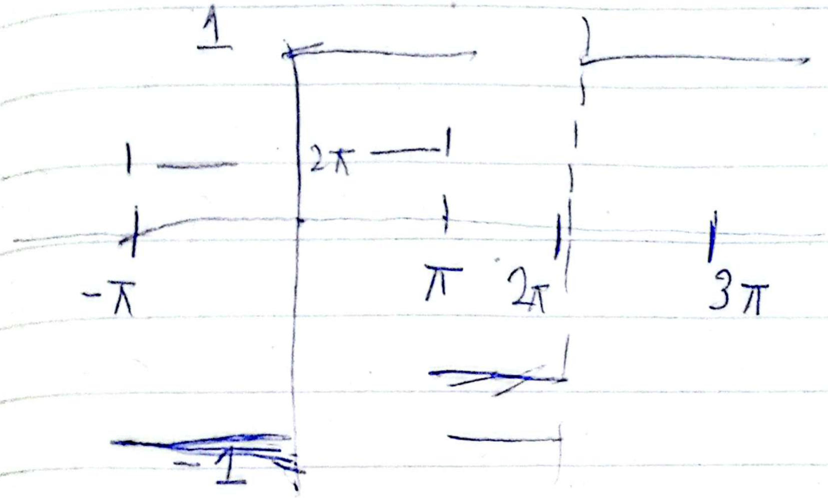
$T_n = a + (n-1)d$

~~1+(n~~ $= 2n-1$

$1 + (n-1)2$

$1 + 2n - 2$

$2n - 1$



1
2π
-π
π
2π
3π
-1

Tuesday

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* Fourier Series is also used to find pi

Let $x = \pi/2$

$$1 = \frac{4}{\pi}\left[\sin(\pi/2) + \frac{\sin 3(\pi/2)}{3} + \frac{\sin 5(\pi/2)}{5}\right]$$

$$1 = \frac{4}{\pi}\left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots\right]$$

$$\Rightarrow \pi = 4\left(1 - 1/3 + \frac{1}{5} - \frac{1}{7} + \dots\right)$$

$$\pi \cong 4\left(1 - \frac{1}{3} + \frac{1}{5}\right) \cong 3.46$$

Ex 11.1

Qs 12-21

* Qs 12 F(x) in prob 6. Even Function

$F(-x) = F(x)$

$F(x) = |x|$; $-\pi < x < \pi$; $f(x+2\pi) = f(x)$

* $F(-x) = -f(x)$ Odd function

OR

$$f(x) = \begin{cases} -x & ; -\pi < x \leq 0 \\ x & ; 0 < x < \pi \end{cases}$$

Sol:-

Req F. Series is:-

$$F(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \rightarrow (1)$$

Here

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(x)\, dx$$

$$= \frac{1}{2\pi} \left[\int_{-\pi}^{0} -x\, dx + \int_{0}^{\pi} x\, dx \right]$$

$$= \frac{1}{2\pi} \left[\frac{-x^2}{2} \Big|_{-\pi}^{0} + \frac{x^2}{2} \Big|_{0}^{\pi} \right]$$

$$= \frac{1}{2\pi} \left[-\frac{1}{2} \left[0 - \pi^2 \right] + \frac{1}{2} \left[\pi^2 \right] \right]$$

$$= \frac{1}{2\pi} \left[\frac{\pi^2}{2} + \frac{\pi^2}{2} \right]$$

$$= \frac{1}{2\cancel{\pi}} \left[\frac{\cancel{2}\pi^2}{\cancel{2}} \right]$$

$$\Rightarrow \quad \boxed{a_0 = \frac{\pi}{2}}$$

Note:-

* If 'f' is an even function

$$\int_{-a}^{a} F = 2\int_{0}^{a} F$$

* If 'f' is an odd function

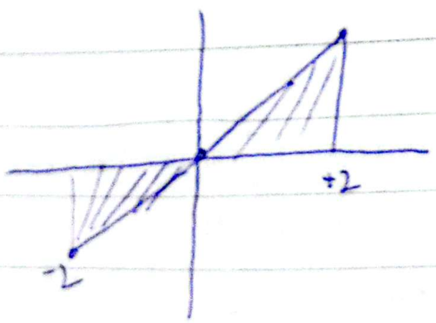
$$\int_{-a}^{a} f = 0$$

Eg here:

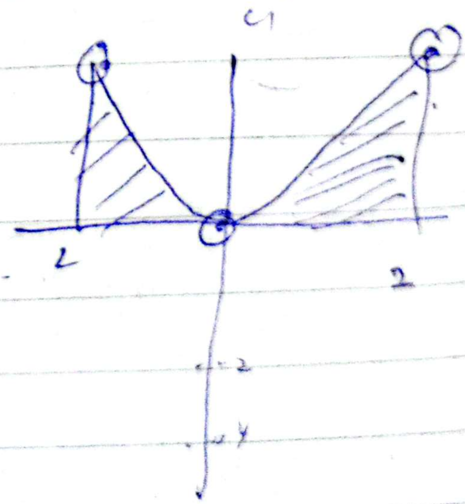
$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |x| \, dx = \frac{1}{\cancel{2}\pi}\,\cancel{2}\int_{0}^{\pi} x \, dx$$

$$= \frac{1}{\pi}\frac{x^2}{2}\Big|_{0}^{\pi} = \frac{1}{\cancel{\pi}}\left(\frac{\pi^{\cancel{2}}}{2}\right) = \pi/2$$

$f(x) = x \; ; \; -2 \le x \le 2$



$F(x) = x^2 \; ; \; -2 \le x \le 2$



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Note:-

Even x Even = Even

$x^2 \cdot x^6 = x^8$

odd x odd = Even

$x^3 \cdot x^5 = x^8$

Even x odd = odd

$x^2 \times x^3 = x^5$

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$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} F(x)\cos nx\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi} |x|\cos nx\,dx$$

$$= \frac{2}{\pi}\int_{0}^{\pi} x\cos nx\,dx \quad ; \quad \int_{-a}^{a} \underset{\text{Even}}{f} = 2\int_{0}^{a} F$$

$$= \frac{2}{\pi}\left[\left.\frac{x(\sin nx)}{n}\right|_0^{\pi} - \int_0^{\pi} (1)\frac{\sin nx}{n}\,dx \right]$$

$$= \frac{2}{\pi}\left[\left.\frac{\cos nx}{n^2}\right|_0^{\pi} \right]$$

$$= \frac{2}{\pi n^2}\left\{ \cos n\pi - \cos 0 \right\}$$

$$\boxed{a_n = \frac{2}{\pi n^2}\left\{(-1)^n - 1\right\}}$$

Note.

$\cos n\pi = (-1)^n$

$\cos \pi = -1$

$\cos 2\pi = 1$

$\cos 3\pi = -1$

$\cos 4\pi = 1$

$\because \sin n\pi = 0$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \underset{\text{Even}}{|x|}\,\underset{\text{odd}}{\sin nx}\,dx = 0 \qquad \because \int_{-a}^{a} f = 0$$

If 'f' is odd

$$eq\,n\,(1) \Rightarrow f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi}\left\{(-1)^n - 1\right\}\cos nx$$

$$+ 0 \sin nx$$

$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

$$\boxed{|x| = \frac{\pi}{2} + \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{\{(-1)^n - 1\}\cos nx}{n^2}} \text{ Ans}$$

$$|x| = \frac{\pi}{2} - \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{\cos(2n-1)x}{(2n-1)^2}, \quad 1, 3, 5, 7 \ldots \quad 2n-1$$

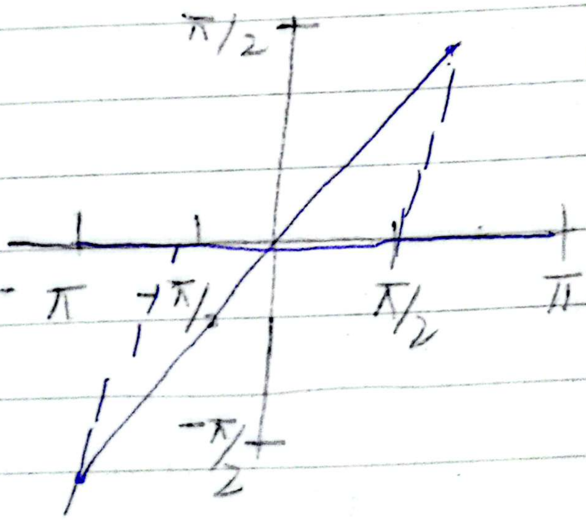
Q.15 $F(x) = x^2; \; 0 < x < 2\pi; \; f(x+2\pi) = F(x)$

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} F(x)\,dx, \quad a_n = \frac{1}{\pi}\int_0^{2\pi} F(x)\cos nx\,dx$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} F(x)\sin nx\,dx$$

Note: Complete yourself.

Q. 16



Complete yourself

$y = -x + 0$

$$F(x) = \begin{cases} 0 & -\pi < x < -\pi/2 \\ \cdot & -\pi/2 < x < \pi/2 \\ 0 & \pi/2 < x < 0 \end{cases}$$

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$\cos n\pi = (-1)^n$; $\sin n\pi = 0$

$\sin n\frac{\pi}{2} = ?$

$1, 0, -1, 0, 1, 0, -1, 0, \ldots$

$\sin\frac{\pi}{2} = 1$, $\sin\frac{\cancel{2}\pi}{\cancel{2}} = 0$

$\sin\frac{3\pi}{2} = -1$, $\sin\frac{\overset{2}{\cancel{4}}\pi}{\cancel{2}} = 0$

$\sin\frac{5\pi}{2} = 1$, $\sin\frac{\overset{3}{\cancel{6}}\pi}{\cancel{2}} = 0$

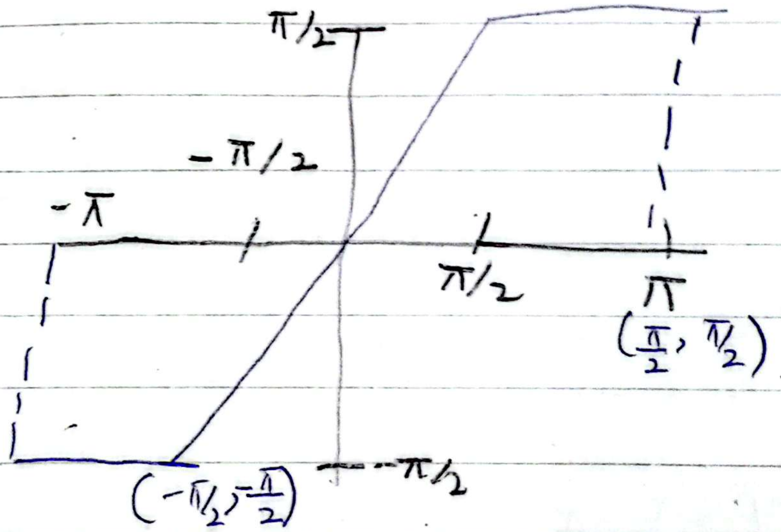
$\sin\frac{7\pi}{2} = -1$, $\sin\frac{\overset{4}{\cancel{8}}\pi}{\cancel{2}} = 0$

$1, 3, 5, 7$ | $2, 4, 6, 8$

$T_n = 2n-1$ | $T_n = 2n \to 0$

$(-1)^{n+1}$

Q.20



$$f(x) = \begin{cases} -\pi/2 \,; & -\pi < x < -\frac{\pi}{2} \\ x \,; & -\pi/2 < x < \pi/2 \\ \pi/2 \,; & \pi/2 < x < \pi \end{cases} \quad ; f(x+2\pi) = f(x)$$

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

$$y - \left(\frac{-\pi}{2}\right) = \frac{\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)}{\frac{\pi}{2} - \left(\frac{-\pi}{2}\right)}\left(x - \left(-\frac{\pi}{2}\right)\right)$$

$$y + \cancel{\frac{\pi}{2}} = \frac{\cancel{2\pi}/\cancel{2}}{\cancel{2\pi}/\cancel{2}}\, x + \cancel{\frac{\pi}{2}} \qquad y = x$$

Req F. Series is

$$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx) \longrightarrow (1)$$

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} F(x)\,dx = \frac{1}{2\pi}\left\{\int_{-\pi}^{-\pi/2} \frac{-\pi}{2}\,dx + \int_{-\pi/2}^{\pi/2} x\,dx + \int_{\pi/2}^{\pi} \frac{\pi}{2}\,dx\right\}$$

$$a_0 = \frac{1}{2\pi}\left\{\frac{-\pi}{2}\int_{-\pi}^{-\pi/2} dx + 0 + \frac{\pi}{2}\int_{\pi/2}^{\pi} dx\right\}; \int_{-a}^{a} f = 0; \text{ if } f \text{ is odd}$$

$$= \frac{1}{2\pi}\left\{\frac{-\pi}{2}\, x\Big|_{-\pi}^{-\pi/2} + \frac{\pi}{2}\, x\Big|_{\pi/2}^{\pi}\right\}$$

$$= \frac{1}{2\pi}\left\{\frac{-\pi}{2}\left(\frac{-\pi}{2} + \pi\right) + \frac{\pi}{2}\left(\pi - \frac{\pi}{2}\right)\right\}$$

$$= \frac{1}{2\cancel{\pi}} \cdot \frac{\cancel{\pi}}{2}\left\{\cancel{\frac{\pi}{2}} - \cancel{\pi} + \cancel{\pi} - \cancel{\pi/2}\right\} = 0$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} F(x)\cos nx\,dx = 0; \text{ Even} \times \text{odd} = \text{odd}$$

$$\int_{-a}^{a} f = 0; f \text{ is odd}$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \overset{\text{odd}}{F(x)}\, \overset{\text{odd}}{\sin nx}\, dx = \frac{1}{\pi}\, 2\int_0^{\pi} F(x)\sin nx\, dx$$

$$= \frac{2}{\pi}\left\{ \int_0^{\pi/2} x\sin nx\, dx + \int_{\pi/2}^{\pi} \frac{\pi}{2}\sin nx\, dx \right\}$$

$$= \frac{2}{\pi}\left[\left\{ x\frac{(-\cos nx)}{n}\Big|_0^{\pi/2} - \int_0^{\pi/2} (1)\frac{(-\cos nx)}{n}dx \right\} + \frac{\pi}{2}\int_{\pi/2}^{\pi}\sin nx\, dx\right]$$

$$= \frac{2}{\pi}\left[\frac{1}{n}\left\{-\frac{\pi}{2}\cos\frac{n\pi}{2} - 0\right\} + \frac{\sin nx}{n^2}\Big|_0^{\pi/2} + \frac{\pi}{2}\left(\frac{-\cos nx}{n}\right)\Big|_{\pi/2}^{\pi}\right]$$

$$= \frac{2}{\pi}\left[-\frac{\pi}{2n}\cos\frac{n\pi}{2} + \frac{1}{n^2}\left(\sin\frac{n\pi}{2}\right) - \frac{\pi}{2n}\left\{\cos n\pi - \cos\frac{n\pi}{2}\right\}\right]$$

$$= \frac{2}{\pi}\left\{ \frac{\sin\frac{n\pi}{2}}{n^2} - \frac{\pi}{2n}(-1)^n \right\}$$

Req. F. Series is

$$\boxed{F(x) = 0 + \sum_{n=1}^{\infty} 0 + \frac{2}{\pi}\left\{\frac{\sin n\frac{\pi}{2}}{n^2} - \frac{\pi}{2n}(-1)^n\right\}\sin nx}$$

$$= \frac{2}{\pi}\left[\sum_{n=1}^{\infty}\frac{\sin\frac{n\pi}{2}}{n^2}\sin nx - \frac{\pi}{2}\sum_{n=1}^{\infty}\frac{(-1)^n \sin nx}{n}\right]$$

$\sin(2n-1)\frac{\pi}{2}$

$$= \frac{2}{\pi}\left\{\sum_{n=1}^{\infty} \frac{\sin(2n-1)\pi/2 \, \sin(2n-1)x}{(2n-1)^2} \cdot \frac{\pi}{2}\right.$$

$$\left. - \frac{\pi}{2}\sum_{n=1}^{\infty} \frac{(-1)^n \sin nx}{n}\right\}$$

$$= \frac{2}{\pi}\left\{\sum_{n=1}^{\infty} \frac{(-1)^{n+1} \sin(2n-1)x}{(2n-1)^2} - \frac{\pi}{2}\sum_{n=1}^{\infty}\frac{(-1)^n \sin nx}{n}\right\}$$

$$F(x) = \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{(-1)^{n+1}\sin(2n-1)x}{(2n-1)^2} - \sum_{n=1}^{\infty}\frac{(-1)^n \sin nx}{n}$$

G. Fourier Series

* If $F(x)$ is periodic function of arbitrary period "2L":

$$F(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}$$

$$a_0 = \frac{1}{2L}\int_{-L}^{L} F(x)\,dx \; ; \; a_n = \frac{1}{L}\int_{-L}^{L} F(x)\cos\frac{n\pi x}{L}\,dx$$

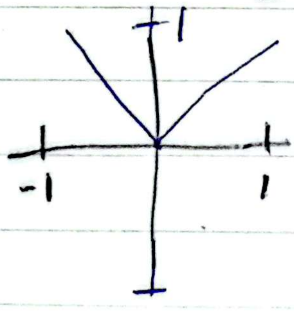
$$b_n = \frac{1}{L}\int_{-L}^{L} F(x)\sin\frac{n\pi x}{L}\,dx$$

$$\cancel{2}L = \cancel{2}\pi \Rightarrow L = \pi$$

$$F(x) = a_0 + \sum_{n=1} a_n \cos nx + b_n \sin nx \; ; \; a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} F(x)\,dx \; ; \; a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} F(x)\cos nx \; ; \; b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

X — X

Ex 11.2 Qs 8-17



$$f(x) = |x| \; ; \; -1 < x < 1 \text{ or } F(x) = \begin{cases} -x \; ; & -1 < x < 0 \\ x \; ; & 0 < x < 1 \end{cases}$$

$$2L = 2$$

$$\boxed{L = 1}$$

$$a_0 = \frac{1}{2L}\int_{-L}^{L} F(x)\,dx = \frac{1}{2}\int_{-1}^{1}|x|\,dx = \frac{1}{\cancel{2}}\cdot\cancel{2}\int_{0}^{1} x\,dx$$

$$= \frac{x^2}{2}\Big|_0^1 \Rightarrow \frac{1}{2}$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx = \frac{1}{1}\int_{-1}^{1}|x|\sin n\pi x\,dx = 0$$

$$a_n = \frac{1}{L}\int_{-L}^{L} F(x)\cos\frac{n\pi x}{L} = \frac{1}{1}\int_{-1}^{1}|x|\cos\frac{n\pi x}{1}\,dx$$

$$= 2\int_{0}^{1} x\cos n\pi x\,dx$$

Week 12
Tuesday
CVF

Ex 11.2

Q.11 $F(x) = x^2; \quad -1 < x < 1; F(x+2) = F(x)$

$2L = 2, \boxed{L = 1}$

Req F. Series is

$$F(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right) \quad \text{—} (1)$$

Then

$$a_0 = \frac{1}{2L}\int_{-L}^{L} F(x)\,dx = \frac{1}{2}\int_{-1}^{1} x^2 dx = \frac{1}{\cancel{2}} \cdot \cancel{2}\int_0^1 x^2 dx = \boxed{\frac{1}{3}}$$

$$a_n = \frac{1}{L}\int_{-L}^{L} F(x) \cos\frac{n\pi x}{L}$$

$$= \frac{1}{1}\int_{-1}^{1} x^2 \cos\frac{n\pi x}{1}$$

$$= 2\int_0^1 x^2 \cos n\pi x\, dx$$

$$\because \int uv = u\int v - \int u' \int v$$

$$= 2\left[\frac{x^2 \sin n\pi x}{n\pi}\Big|_0^1 - \int_0^1 2x\left(\frac{\sin n\pi x}{n\pi}\right) dx\right]$$

Note : In half range extension (series, one of the coefficients is zero).

If function is even = Cosine Series

If function is odd = Sin Series

If neither = both will be there.

$$= \frac{-4}{n\pi}\int_0^1 x \sin n\pi x \, dx$$

$$= \frac{-4}{n\pi}\left[\frac{x(-\cos n\pi x)}{n\pi}\Big|_0^1 - \int_0^1 (1)\left(\frac{-\cos n\pi x}{n\pi}\right)\right]$$

$$= \frac{-4}{n\pi}\left[\frac{-(1)\cos n\pi + 0}{n\pi} + \frac{1}{n\pi}\frac{\overset{\nearrow 0}{\sin n\pi x}}{n\pi}\Big|_0^1\right]$$

$$a_n = \frac{4}{n^2\pi^2}(-1)^n \; ; \quad \because \cos n\pi = (-1)^n$$

$$b_n = \frac{1}{L}\int_{-L}^{L} F(x)\sin n\pi x \, dx = \frac{1}{1}\int_{-1}^{1} x^2 \sin \frac{n\pi x}{1} dx$$

$$= 0 \;, \quad \because \int_{-a}^{a} F \doteq 0 \; ; \text{ If } f \text{ is an odd function}$$

Putting the values in eqn ①

$$\text{eq ①} \Rightarrow F(x) = \frac{1}{3} + \sum_{n=1}^{\infty}\left(\frac{4}{n^2\pi^2}(-1)^n \cos\frac{n\pi x}{1} + (0)\sin\ldots\right)$$

$$\boxed{x^2 = \frac{1}{3} + \frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n \cos n\pi x}{n^2}}$$

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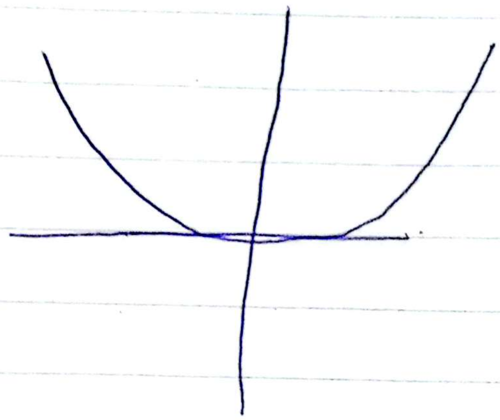
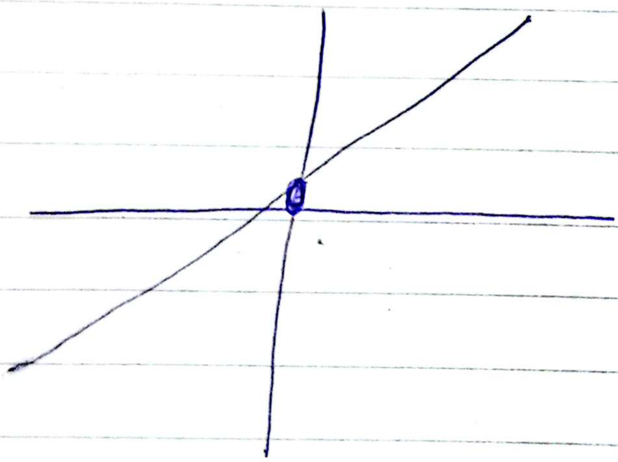
Qs 8 - 17

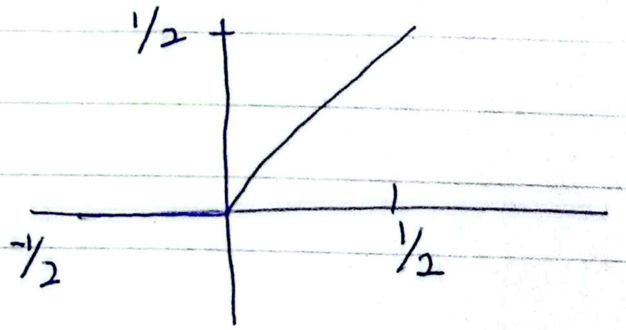
* Even



* Odd



* Neither



Thursday

23-04-2026 Online

* COMPLEX VALUED FUNCTION OR COMPLEX FUNCTION

~~f = u + iv~~ $f : \mathbb{C} \rightarrow \mathbb{C}$

$F(z) = u(x,y) + iv(x,y) = u + iv$

* EXAMPLE :-

$f : \mathbb{C} \rightarrow \mathbb{C}$

$F(z) = z^2$

$\Rightarrow f(z) = (x+iy)^2$

$f(z) = \underbrace{(x^2-y^2)}_{u} + i\underbrace{2xy}_{v}$

$u + iv$

* ZEROS AND POLES OF COMPLEX FUNCTION

If $f(z)$ is a complex valued function then

(i) $Z = Z_0$ is ~~caused~~ called zero of 'f' if $f(Z_0) = 0$

(ii) $Z = Z_0$ is called a pole, or singularity of this function if $F(Z_0) = \infty$

Part D (16.2)

* Find all zeroes of corresponding poles of $F(z) = \dfrac{(2z-3)(z-1)}{(z-2)(z+4)}$

* For Zeroes:-

$F(z) = 0$

$\Rightarrow \dfrac{(2z-3)(z-1)}{(z-2)(z+4)} = 0$

$\Rightarrow (2z-3)(z-1) = 0$

$2z-3=0 \ , \ z-1$

$\boxed{z = \frac{3}{2}} \ , \ \boxed{z = 1}$

* FOR POLES:-

$F(z) = \infty$

$\dfrac{(2z-3)(z-1)}{(z-2)(z+4)} = \dfrac{1}{0}$

$0 = (z-2)(z+4)$

$\boxed{z=2} \ , \ \boxed{z=-4}$

x—x

13.4 Cauchy-Riemann Equations

Suppose $F(z)$ is a complex valued function i.e, $F(z) = u + iv$ then

$z^2 = (x)^2 - 2(x)(iy) + (iy)^2$
$x^2 - y^2$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

and

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Cauchy Rieman equation (CR-equation)

* ANALYTIC FUNCTION:

complex differentiable

Holomorphic

DEF:

IF $f(z)$ is an analytic function then it satisfies CR equation.

* Show that the given function is an analytic function

$$f : \mathbb{C} \to \mathbb{C}$$
$$f(z) = z^2$$

PROOF:-

$$f(z) = z^2 = \underbrace{(x^2 - y^2)}_{u} + i\underbrace{2xy}_{v}$$

Here

$$u = x^2 - y^2, \quad v = 2xy$$

$$\frac{\partial u}{\partial x} = 2x \ , \ \frac{\partial u}{\partial y} = -2y$$

$$\frac{\partial v}{\partial x} = 2y \ , \ \frac{\partial v}{\partial y} = 2x$$

$$\left.\begin{array}{l} \because \dfrac{\partial u}{\partial x} = \dfrac{\partial v}{\partial y} \\ \text{and } \dfrac{\partial u}{\partial y} = -\dfrac{\partial v}{\partial x} \end{array}\right\}$$

$\therefore$ $F(z) = z^2$ is an analytic function.

x — x

- Identify whether or not $F(z) = \bar{z}^2$ is an analytic Function

Sol:-

$$F(z) = (\bar{z})^2 = (x - iy)^2 = \overbrace{(x^2 - y^2)}^{u} + i\overbrace{(-2xy)}^{v}$$

Here

$$u = x^2 - y^2 \text{ and } v = -2xy$$

$$\frac{\partial u}{\partial x} = 2x \ , \ \frac{\partial u}{\partial y} = -2y$$

$$\frac{\partial v}{\partial x} = 2y \ , \ \frac{\partial v}{\partial y} = -2x$$

$$\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$$

and

$$\frac{\partial u}{\partial y} \neq -\frac{\partial v}{\partial y}$$

$\therefore$ Not an analytic function

Ex # 13.4

Q:4 $F(z) = e^{x}(\text{Cos}y - i\text{Sin}y)$

$= e^{x}\text{Cos}y + i(-e^{x}\text{Sin}y)$

Here

$u = e^{x}\text{Cos}y \quad , \quad v = -e^{x}\text{Sin}y$

$\frac{\partial u}{\partial x} = e^{x}\text{Cos}y \quad , \quad \frac{\partial u}{\partial y} = -e^{x}\text{Sin}y$

$\frac{\partial v}{\partial x} = -e^{x}\text{Sin}y \quad , \quad \frac{\partial v}{\partial y} = -e^{x}\text{Cos}y$

$\because \frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y} \quad , \quad \frac{\partial u}{\partial y} \neq \frac{\partial v}{\partial y}$

$\therefore$ Function is not an analytic Function

Note:

$z = x + iy = r(\text{Cos}\,\theta + i\text{Sin}\,\theta)$

$\bar{z} = x - iy) = r(\text{Cos}\,\theta - i\text{Sin}\,\theta)$

$\bar{z}$ is not an analytic function.

Week 13 () → Uncountable

Tuesday

CVFA

* Z - TRANSFORMATION

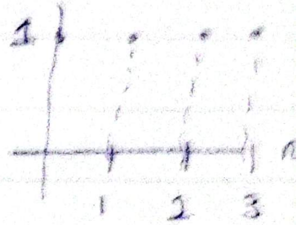
* Two sides Z-transform is defined as:

$$F(Z) = \sum_{n=-\infty}^{\infty} f[n] Z^{-n}$$; where n is an integer and Z is a complex number.

* One-sided Z-transform $= \sum_{n=0}^{\infty} F[n] Z^{-n}$ → Countable

---

$$u[n] = \begin{cases} 1 & ; n \geq 0 \\ 0 & ; e.w \end{cases}$$



$$\mathcal{Z}\{u[n]\} = \sum_{n=0}^{\infty} u[n] Z^{-n}$$

Infinite Geometric Series

$$= \sum_{n=0}^{\infty} (1) Z^{-n} = 1 + Z^{-1} + Z^{-2} + \dots = \frac{1}{1-\frac{1}{Z}}$$

$$= \frac{Z}{Z-1} ; |Z| < 1$$

* EXAMPLE # 2 :-

$$f[n] = \begin{cases} a & ; n \geq 0 ; a \text{ is a constant} \\ 0 & ; e.w \end{cases}$$

$$\mathcal{Z}(f[n]) = \sum_{n=0}^{\infty} f[n] Z^{-n}$$

$$= \sum_{n=0}^{\infty} a Z^{-n} \Rightarrow a \sum_{n=0}^{\infty} Z^{-n}$$

$$= a\left(\frac{1}{1-Z}\right) = a\left(\frac{Z}{Z-1}\right)$$

* Example # 3:-

$$F[n] = \begin{cases} a^n ; & n \geq 0 \\ 0 ; & e.w \end{cases}$$

$$\mathcal{Z}(F[n]) = \sum_{n=0}^{\infty} F[n] z^{-n} \Rightarrow \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{z}{a}\right)^{-n}$$

$$= a \sum_{n=0}^{\infty} z^{-n} = a\left[1 + z^{-1} + z^{-2} \ldots\right] = a\left[1 + \frac{1}{z} + \frac{1}{z^2} \ldots\right.$$

Week 13

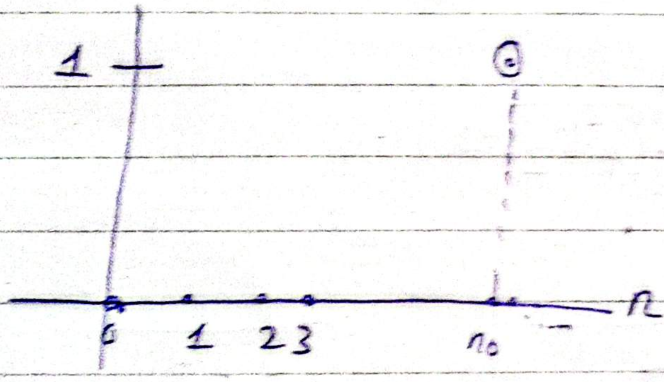
CVF

30-04-2026 Online (Thursday)

$$\delta(n-n_0) = \begin{cases} 1 & ; \; n = n_0 \\ 0 & ; \; n \neq n_0 \end{cases}$$

* Impulse Function:-

$$F[n] = \delta(n-n_0)$$



$$\mathcal{Z}(\delta(n-n_0)) = \sum_{n=0}^{\infty} \delta(n-n_0) Z^{-n} = 0+0+0 + \dots (1) Z^{-n_0} + 0+0$$

$$= Z^{-n_0} \; ; \; Z \in \mathbb{C}$$

* EXAMPLE:-

$$F[n] = e^{an} \quad \text{or} \quad u(n) e^{an}$$

$$\mathcal{Z}(F[n]) = \sum_{n=0}^{\infty} f[n] Z^{-n}$$

$$= \sum_{n=0}^{\infty} u(n) e^{an} Z^{-n}$$

$$= \sum_{n=0}^{\infty} (e^{-a})^{-n} Z^{-n}$$

$$= \sum_{n=0}^{\infty} (e^{-a} z)^{-n}$$

$$= (e^{-a} z)^{0} + (e^{-a} z)^{-1} + (e^{-a} z)^{-2} + (e^{-a} z)^{-3} + \ldots$$

$$\boxed{\begin{array}{l} \text{Common ratio} = (e^{-a} z)^{-1} \\ \dfrac{a}{1-r} \end{array}}$$

$$= \frac{1}{1-(e^{-a} z)^{-1}}$$

$$= \frac{1}{1-\frac{1}{e^{-a} z}} \Rightarrow \frac{1}{1-\frac{e^{a}}{z}}$$

$$\boxed{Z(e^{an}) = \frac{z}{z-e^{a}}} = \text{Ans}$$

Ans for $Z(e^{2n}) = \dfrac{z}{z-e^{2}}$

EXAMPLE :-

$f[n] = n$

$$\mathcal{Z}(f[n]) = \sum_{n=0}^{\infty} f[n] z^{-n} = \mathcal{Z}(n)$$

$$\because \mathcal{Z}(1) = \sum_{n=0}^{\infty} (1) z^{-n}$$

$$\frac{(1)z}{z-1} = \sum_{n=0}^{\infty} z^{-n} \; ; \quad \because \mathcal{Z}(a) = \frac{az}{z-1}$$

$$\frac{z}{z-1} = \sum_{n=0}^{\infty} z^{-n}$$

Diff w.r.t z

$$\frac{d}{dz}\left(\frac{z}{z-1}\right) = \frac{d}{dz} \sum_{n=0}^{\infty} z^{-n}$$

$$\frac{(z-1)(1) - (z)(1)}{(z-1)^2} = \sum_{n=0}^{\infty} \frac{d}{dz}\left(z^{-n}\right)$$

$$\Rightarrow \frac{\cancel{z} - 1 - \cancel{z}}{(z-1)^2} = \sum_{n=0}^{\infty} -n z^{-n-1}$$

$$\Rightarrow \frac{\cancel{-}1}{(z-1)^2} = \cancel{-}\sum_{n=0} n z^{-n} z^{-1}$$

$$\frac{1}{(z-1)^2} = z^{-1}\sum_{n=0}^{\infty} n z^{-n}$$

$$\Rightarrow \frac{z}{(z-1)^2} = \sum_{n=0}^{\infty} n z^{-n}$$

$$\therefore \boxed{\mathcal{Z}(n) = \frac{z}{(z-1)^2}}$$

x ___ x

* $\mathcal{Z}(n^2) = ??$

$$\because \mathcal{Z}(n) = \frac{z}{(z-1)^2}$$

$$\sum_{n=0}^{\infty} n z^{-n} = \frac{z}{(z-1)^2}$$

Ans $\sum_{n=0}^{\infty} n^2 z^{-n} = \mathcal{Z}(n^2) =$

$$= \frac{z(z+1)}{(z-1)^3}$$

diff w.r.t z

$$n \frac{d}{dz} \sum_{n=0}^{\infty} z^{-n} = \frac{d}{dz} \frac{z}{(z-1)^2}$$

$$n \sum_{n=0}^{\infty} \frac{d}{dz} z^{-n} = \frac{(z-1)^2 (1) - (z) \frac{d}{dz} (z-1)^2}{(z-1)^4}$$

$$n \sum_{n=0}^{\infty} -n z^{-n} = \frac{(z-1)^2 - z\{2(z-1)(1)\}}{(z-1)^4}$$

$$-\sum n^2 z^{-n} = (z-1)\frac{[z-1-2z^2]}{(z-1)^4}$$

$$-\sum n^2 z^{-n} = \cancel{\frac{-z-1-2z^2}{(z-1)^3}}$$

$$\sum n^2 z^{-n} = \frac{z(z+1)}{(z-1)^3}$$

x — x

Example:-

$$f[n] = u(n)\,\text{Cos}\,bn$$

$$z(f[n]) = \sum_{n=0}^{\infty} f[n]\,z^{-n}$$

$$z(u(n)\,\text{Cos}\,bn) = \sum_{n=0}^{\infty} u(n)\,\text{Cos}(bn)\,z^{-n}$$

$$= \sum_{n=0}^{\infty} \text{Cos}(bn)\,z^{-n}$$

$$= \sum_{n=0}^{\infty} \frac{e^{i(bn)} + e^{-i(bn)}}{2}\,z^{-n}$$

$$z(u(n)\,\text{Cos}(bn)) = \frac{1}{2}\left[\sum_{n=0}^{\infty} e^{ibn} z^{-n} + \sum_{n=0}^{\infty} e^{-ibn} z^{-n}\right]$$

$$= \frac{1}{2}\left[\frac{z}{z-e^{ib}} + \frac{z}{z-e^{-ib}}\right]$$

$$\because z(e^{an}) = \sum_{n=0}^{\infty} e^{an} z^{-n} = \frac{z}{z-e^{a}}$$

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$$= \frac{1}{2}\left[\frac{z\left(z-e^{-ib}\right)+z\left(z-e^{ib}\right)}{\left(z-e^{ib}\right)\left(z-e^{-ib}\right)}\right]$$

$$= \frac{z^2 - z\cos b}{z^2 - 2z\cos b + 1}, \quad \cos b = \frac{e^{ib}+e^{-ib}}{2}$$

14th WEEK

TUESDAY

05-05-2026

EXAMPLE:-

$$f[n] = na^{n+2}\, u(n) + 2\delta(n)$$

$$\mathcal{Z}(F[n]) = \mathcal{Z}(na^{n+3} u(n)) + 2\mathcal{Z}(\delta($$

$$= a^2 \mathcal{Z}(n a^n u(n)) + 2\mathcal{Z}(\delta(n \quad \text{—(1)}$$

$$\because \mathcal{Z}(n) = \frac{z}{(z-1)^2}$$

$$\therefore \mathcal{Z}(a^n n) = \frac{z/a}{\left(\frac{z}{a} - 1\right)^2}$$

$$= \frac{z/a}{\frac{(z-a)^2}{a^4}}$$

$$= \frac{az}{(z-a)^2}$$

Putting in eq (1)

$$\text{Eq } ① \Rightarrow \mathcal{Z}[f[n]] = a^2 \frac{az}{(z-a)^2} + 2(1)$$

$$= \frac{a^3 z}{(z-a)^2} + 2$$

x — x

***EXAMPLE:-**

$$F(z) = \frac{z}{z^2+7z+10}$$

$$\Rightarrow \mathcal{Z}(f[n]) = \frac{z}{z^2+7z+10}$$

$$\Rightarrow f[n] = \mathcal{Z}^{-1}\left(\frac{z}{z^2+7z+10}\right)$$

$$\Rightarrow f[n] = \mathcal{Z}^{-1}\left(\frac{z}{(z+5)(z+2)}\right) \rightarrow ①$$

Consider:-

$$\frac{1}{(z+5)(z+2)} = \frac{A}{(z+5)} + \frac{B}{(z+2)}$$

3 questions (20)

1 from Z-transform (Zero + Poles, Equation)

1 from Laplace transform

1 from Laplace, Fourier, Infinite

$$= \frac{1}{3(z+2)} - \frac{1}{3(z+5)}$$

Putting in Eq 1

$$F[n] = 3^{-1}\left[\frac{z}{3(z+2)} - \frac{z}{3(z+5)}\right]$$

$$= \frac{1}{3} 3^{-1}\left(\frac{z}{z-(-2)}\right) - \frac{1}{3} z^{-1}\left(\frac{z}{z-(-5}\right.$$

$$= \frac{1}{3}(-2)^n - \frac{1}{3}(-5)^n; \quad \because 3(a^n) = \frac{z}{z-}$$

$$\rightarrow a^n = 3^{-1}\left(\frac{}{z-}\right.$$